MILITARY SPECIFICATION

HORSERADISH, PREPARED, CHILLED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers prepared, chilled horseradish for use by the Armed Forces as an item of general issue. This product has a limited storage stability.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-G-460 - Glass Containers, One Gallon Capacity and Smaller, For Other Than Medicinal Products; Packaging and Packing Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-668 - Sanitary Standards For Food Plants.

FSC 8950

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Agricultural Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material.- All materials shall be edible, clean, and sound.

* 3.1.1 Horseradish root.- Horseradish root (A. lapathifolia Gilib) shall be fresh, sound, and of good grade.

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3.1.2 Salt.- Salt shall be white, refined, sodium chloride with or without anticaking agent. Iodized salt shall not be used.

* 3.1.3 Distilled vinegar.- Distilled vinegar shall be spirit, distilled, or grain vinegar made by the acetous fermentation of dilute distilled alcohol. It shall be clear, bright, transparent, and free from any pronounced odor other than that of acetic acid.

* 3.1.4 Preservative.- Preservative shall be sodium sulfite or sodium metabisulfite.

* 3.2 Preparation.- The product shall be prepared from fresh horseradish roots (A. lapathifolia Gilib), vinegar, and salt. The root material shall be ground or grated with or without water of the brine formulation. The ground root shall be mixed in such manner as to assure wetting by the brine (vinegar, salt and water) as required.

* 3.3 Finished product.- The finished product shall have a white to off-white color, a typical pungent aroma, and characteristic bite. It shall be free from skin, slivers, root hairs, stalks, coloring and preservative materials other than sodium sulfite and sodium metabisulfite. Corn starch, turnips, parsnips, radishes, or other vegetable materials shall not be added. In addition, the product shall comply with the following requirements:

Total solids	Not less than 16 percent
Titratable acidity (as acetic)	Not more than 3 percent nor less than 2.5 percent
Ash	Not more than 3 percent
Sulfur dioxide	Not less than 50 or more than 100 ppm.
Volatile oil (allyl isothiocyanate)	Not more than 0.07 percent
Salt	Not more than 2.25 percent

3.3.1 The packaged product shall be stored at a refrigerated temperature (32° - 38°F.).

* 3.4 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

* 3.5 Workmanship.- The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-668.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Preacceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.1.1 Ingredients.- Conformance of ingredients to identity requirements (kinds) shall be ascertained by visual examination; examination of labels, invoices, or other valid documents, as applicable. In addition, vinegar and salt shall be examined organoleptically, as necessary; to determine conformance to 3.1 and condition requirements as may be contained in the pertinent ingredient paragraph. The sample unit shall be one pound of salt and one pint of vinegar, each extracted from a primary container. The sample size shall be five. In the event a lot consists of less than 5 containers, a sample shall be extracted from each container in the lot. Nonconformance to identity requirements or one or more sample units not conforming to condition requirements, shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

4.3.2 Examination of end item.- Examinations of the end item shall be in accordance with tables II and III, and 4.3.2.1 and 4.3.2.2. Sampling plans for the aforementioned tables shall be in accordance with table I. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for tables II and IV and percent defective for table III.

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TABLE I.- Sampling plans

Table	Inspection level	Sample unit	Lot size expressed	AQLs Major A	Major B	Minor	Total
II	I	1 filled and closed bottle	Bottles	0.25	1.5	-	4.0
III	S-3	Contents of 1 filled bottle	Bottles	-	-	2.5	-
IV	S-3	4 ounces from a bottle	Bottles	-	1.0	-	-

TABLE II.- Examination of bottle

Categories			Defects
Major A	Major B	Minor	
101			Leaker, broken or cracked.
	151		Checked, chipped, thin spot.
		201	Not clean.
	152		Label missing.
		202	Label insecure.
	153		Nomenclature on label, missing, incorrect or illegible.
		203	Other required label information missing, incorrect or illegible.
	154		Cap loose. 1/
	155		Pitted rust on cap. 2/
	156		Cap liner or sealing compound missing.
	157		Lacquer or enamel on outside of cap and enamel on inside of cap missing, incomplete, or easily abraded with fingertip.

1/ For determining tightness of closure, the following is applicable; condition container to 65 - 75°F. Place the container upright on top of a filled

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and closed fiberboard box. Grasp the sides of the cap firmly with finger tips and turn cap gently counter-clockwise with a minimum of downward pressure for one complete revolution. If cap turns and the container remains stationary, a defect shall be scored.

2/ Rust that can be removed with a soft cloth shall not be considered pitted rust.

TABLE III.- Examination for volume 1/ 2/

Category	Defect
Minor	
201	More than 5 percent under specified volume (for product required to have a volume of not more than 16 fluid ounces).
202	More than 2 percent under specified volume (for product required to have a volume of more than 16 fluid ounces).

1/ Condition container to 65 - 75°F.

2/ Report results to nearest 1/4 ounces for containers required to have a volume of not more than 16 fluid ounces and report results to nearest ounce for containers required to have a volume of more than 16 fluid ounces. Lot shall be rejected if sample data indicates a lot average volume less than specified volume.

TABLE IV.- Examination of product characteristics 1/

Category	Defect
Major B	
151	Product not white to off white.
152	Not free from skin, slivers, root hairs or stalks.
153	Not free from dark particles or other evidence of unsound horseradish.
154	Particles of horseradish not completely wetted.
155	Product does not have a characteristic pungent odor or bite.

1/ Presence of objectionable foreign material (e.g., glass, metal, insects, filth, paint, etc.) shall be cause for rejection of the product. In

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addition, product shall be rejected if label on containers indicated presence of added vegetative material such as cornstarch, turnips, parsnips, and radishes, or added color or preservative (except specified sulfur compounds).

4.3.2.1 Examination for packing, marking, and fiberboard containers.- Examinations shall be in accordance with the quality assurance provisions of PPP-G-460 except that only the defect for marking shall be applicable to level C pack.

4.3.2.2 Storage temperatures.- Conformance to 3.3.1 shall be ascertained by examining the conditions existing in the storage area for the finished product. Product not refrigerated as specified shall be unacceptable.

* 4.3.3 Sampling procedure and acceptance criteria for testing of finished product.- The finished product shall be tested to determine conformance to the analytical requirements specified in 3.3. Procedures for testing shall be in accordance with 4.5. The sample for testing shall be an 8-ounce composite of product derived from the number of primary containers indicated by inspection level S-2. The contents of each primary container shall be mixed thoroughly before extracting the product. Each test result shall be reported to the same unit or decimal place as specified for the specific requirement. The product shall be unacceptable if test results indicate noncompliance with one or more analytical requirements.

* 4.4 Standby test samples.- The Government reserves the right to withdraw and withhold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

* 4.5 Test methods.- Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.5.1 Test procedures.-

* 4.5.1.1 Chemical analysis.- Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists, as indicated below:

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Test	Source	Method
Finished product:		
Total solids	Chapter: Fruits and fruit products.	Moisture in dried fruits.
Titratable acidity (as acetic)	Chapter: Spices and other condiments, section: Vinegars.	Total acids.
Ash	Chapter: Spices and other condiments, section: Spices.	Ash (a).
Volatile oil (allyl isothiocyanate)	Chapter: Spices and other condiments, section: Spices.	Volatile oil in mustard seed.
Salt	Chapter: Meat and meat products, section: Meat.	Salt.
Sulfur dioxide	Chapter: Preservatives and artificial sweeteners, section: Sulfurous acid.	Total sulfurous acid.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging, level C.- The product shall be packaged in commercial glass containers with sealed closures. The closure shall be a metal cap, fitted with a liner or sealing compound suitable for the product. The cap shall be coated overall outside with a corrosion-resistant lacquer or enamel and inside with an enamel furnished by the cap manufacturer in accordance with best commercial practice for the product.

5.2 Packing.- The product shall be packed in accordance with 5.2.1 or 5.2.2 as specified (see 6.1).

* 5.2.1 Level B.- The filled glass containers shall be packed in accordance with the level B requirements of PPP-G-460.

* 5.2.2 Level C.- The product shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation

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rates for such supplies. The shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Labeling and marking.-

* 5.3.1 Glass containers.- The glass containers shall be labeled in accordance with the Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

5.3.2 Shipping containers.- Shipping containers shall be marked in accordance with MIL-STD-129, and as specified in 5.3.2.1.

5.3.2.1 Additional markings.- The following precautionary markings shall be stenciled or printed on the top of the shipping container in letters between 1 and 1-1/2 inches high:

PERISHABLE - KEEP REFRIGERATED
(32°F. to 38°F.)

GLASS - DO NOT DROP OR THROW
or
GLASS - HANDLE WITH CARE

6. NOTES

* 6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing required (see 5.2).

* 6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Project No. 8950-0156

Review activities:

Army - MD
Navy - MC
DSA - SS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Horseradish, Prepared, Chilled MIL-H-35044B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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